



**Verified Carbon
Standard**

10.2MW NAM YEIM HYDRO POWER PROJECT

Project title	10.2MW Nam Yeim Hydro Power Project
Project ID	<i>Verra Project ID</i>
Crediting period	01-Mar-2026 to 28-Feb-2036 (plan)
Original date of issue	05-Dec-2025
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1 PROJECT DETAILS

1.1 Summary Description of the Project

10.2MW Nam Yeim Hydro Power Project (hereafter referred to as the “the project”) is developed by Nam Yeim Hydropower Sole Co., Ltd. in Mok District, Xieng Khouang Province, Lao PDR.

The project is to install a hydro power plant with a total installed capacity of 10.2 MW, provided by two vertical Pelton turbines with a unit capacity of 5.1 MW, a digital SCADA control system, and associated electrical equipment for interconnection to the 110kV grid. The project is anticipated to start operation on 1-Mar-2026 (actual start date to be confirmed at the first verification), with an annual average generation of 35,000 MWh and an annual net generation of 34,650 MWh.

The primary purpose of the project is to generate clean electricity via renewable energy and supply it to national grid of Vietnam, replacing the power generated by fossil fuel plants in the baseline scenario. The project is therefore expected to reduce GHG emissions by an estimation of 22,841 tCO₂e per year, and the total GHG emission reductions are 228,410 tCO₂e during the fixed 10 years crediting period.

The project is not only supplying renewable electricity to grid, but also producing positive environmental and economic benefits and also, promotes the local sustainable development in the following aspects:

- During the construction and operation of the project, it would bring a large number of job opportunities to the local people, especially those from poor families. Then it will help them earn income through working in the project, thus contributing to the economic stability of the local community.
- The roads, bridges, and communication facilities built for the construction of the hydro power plant have significantly improved transportation and communication conditions in the project area, facilitating easier travel for residents and more efficient transport of goods.
- The project provides a stable and reliable electricity supply to meet Vietnam's rapidly growing industrial and residential demand, thereby alleviating the pressure of power shortages.
- By replacing electricity generated from fossil fuels on the Vietnamese grid with renewable power, the project reduces GHG emissions.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
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Validation/ verification	To Be Determined	VCS	To Be Determined	To Be Determined
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1.3 Sectoral Scope and Project Type

Sectoral scope ¹	01- Energy Industries (Renewable/non-renewable sources)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

The project activity conforms to the scope of the VCS Program as defined in section 2.1.1 of the VCS Standard V4.7. The justification for each criterion is provided in the table below:

VCS Scope		Justification
1)	The seven Kyoto Protocol greenhouse gases.	By replacing fossil fuel power generation with hydro power generation, this project reduces emissions of carbon dioxide (CO ₂), which is one of the seven greenhouse gases of the Kyoto Protocol covered under the VCS Program.
2)	Ozone-depleting substances (ODS).	Not applicable. This project does not involve ODS.
3)	Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	Not applicable.
4)	Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	The project uses the CDM-approved methodology AMS-I.D, which is eligible under the VCS Program as it does not appear on the VCS methodology exclusion list. Hence, this criterion is met.
5)	Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable.

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

The project activity is a 10.2 MW small-scale hydropower project located in Lao PDR which is classified as a Least Developed Country (LDC) according to the United Nations' list². The project's GHG emission reductions are achieved by replacing the power generated by fossil fuel plants on the grid, rather than by reducing or destroying emissions generated by the project itself. Therefore, the project meets the requirements of sections 2.1.2 and 2.1.3 of the VCS Standard V4.7.

As per section 3.8.1 of VCS Standard V4.7, Non-AFOLU projects shall complete validation within two years of the project start date. The project is scheduled to start operation on 01-Mar-2026. The project is progressing as planned and is on track to complete its validation before the two-year deadline in order to meet this requirement.

As per para 9 of Tool 20 "Assessment of Debundling for Small-Scale Project Activities (Version 4.1)", a proposed small-scale project activity shall be deemed to be a debundled component of a large project activity if there is a registered small-scale CDM project activity or an application to register another small-scale CDM project activity:

- a) With the same project participants;
- b) In the same project category and technology/measure; and
- c) Registered within the previous 2 years; and
- d) Whose project boundary is within 1 km of the project boundary of the proposed small-scale activity at the closest point.

The project proponent has not registered any other small-scale hydro power projects within the last two years within 1 km of this project's boundary, hence proposed project activity is not a fragmented part of the larger project activity.

1.4.2 AFOLU project eligibility

N/A.

1.4.3 Transfer project eligibility

N/A.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)

² <https://www.un.org/en/conferences/least-developed-countries#:~:text=The%2046%20countries%20currently%20on,%2C%20Kiribati%2C%20Lao%20People's%20Dem.>

☐ Grouped project

1.5.1 Grouped project design

The project activity is not a grouped project. Thus, this section is not applicable.

1.6 Project Proponent

Organization name	SINGAPORE TREASURE CARBON TECH PTE. LTD.
Contact person	Yujia Ren
Title	Project Manager of Technology Department
Address	24 SIN MING LANE, #06-97, MIDVIEW CITY, SINGAPORE 573970
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1.7 Other Entities Involved in the Project

This section will be completed in the full Project Description(PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

Organization name	
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	

1.8 Ownership

The project owner is Nam Yeim Hydropower Sole Co., Ltd., which holds the legal right to control and operate the project activity. The company's legal capacity is evidenced by its business license. Additionally, the business operation license, equipment purchasing contract, and power

purchase agreement collectively demonstrate the ownership of the plant equipment and the right to generate electricity.

The project owner authorizes Singapore Treasure Carbon Tech Pte. Ltd. as the project proponent to develop and manage the VCS project and account.

1.9 Project Start Date

Project start date	01-Mar-2026
Justification	As per section 3.8 of VCS Standard (Version 4.7), the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or carbon dioxide removals. The project is anticipated to start operation on 01-Mar-2026 (actual start date to be confirmed at the first verification) Thus, the project start date is 01-Mar-2026.

1.10 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of fixed crediting period	01-Mar-2026 to 28-Feb-2036 (inclusive of both start and end date) The project is currently under construction. This crediting period is estimated. The actual crediting period will be confirmed during the first verification.

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO₂e/year (project)

☐ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-March-2026 to 31-December-2026	19,149

01-January-2027 to 31-December-2027	22,841
01-January-2028 to 31-December-2028	22,841
01-January-2029 to 31-December-2029	22,841
01-January-2030 to 31-December-2030	22,841
01-January-2031 to 31-December-2031	22,841
01-January-2032 to 31-December-2032	22,841
01-January-2033 to 31-December-2033	22,841
01-January-2034 to 31-December-2034	22,841
01-January-2035 to 31-December-2035	22,841
01-January-2036 to 28-February-2036	3,692
Total estimated ERRs during the fixed crediting period	228,410
Total number of years	10
Average annual ERRs	22,841

1.12 Description of the Project Activity

The project activity is a run-of-river Hydro power project with a total installed capacity of 10.2 MW. The project is promoted by Nam Yeim Hydropower Sole Co., Ltd..

The hydroelectric system for this project mainly consists of turbine-generator sets and a step-up substation. The system comprises two such sets, each composed of a vertical Pelton turbine and a 6300 kVA three-phase synchronous generator. Electricity produced by the generators is integrated into the power grid via the substation equipment. Here, two main transformers are responsible for stepping up the generator output voltage from 6.3 kV to the grid voltage of 110 kV. The station is also fitted with an auxiliary transformer to provide low-voltage electricity for its own use, as well as a diesel generator that serves as an emergency backup power supply.

The Project activity is a new facility (Greenfield). The electricity generated by the project will be transmitted via a dedicated, newly constructed 110 kV transmission line, through the power grid of Lao PDR to the power grid of Vietnam. The project will therefore displace an equivalent

amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid.

Prior to project initiation, the entire electricity delivered to the grid by the project activity, would have otherwise been generated by the operation of grid-connected power plant. The scenario prior to the start of implementation of the project activity is generation of electricity, dominated by thermal power plants, thus leading to GHG emissions. The baseline scenario is the same as the scenario prior to the start of implementation of the project activity.

The project is therefore expected to reduce GHG emissions by an estimation of 22,841 tCO₂e per year, thereon displacing 34,650 MWh/year amount of electricity from the grid over the fixed 10 years crediting period.

Table 1-1: Main Technical Parameters of the project

Pelton turbine parameter							
No	Item	Value	Unit	No	Item	Value	Unit
1	Gross head	382.4	m	6	Runaway speed	1388	rpm
2	Net head	380.41	m	7	Runner diameter	792	mm
3	Turbine	4.65	m	8	Suction head	2.685	m
4	Nominal Turbine	7,873	kW	9	Elevation	535.0	m.a.s.l
5	Nominal speed	750	rpm				
Three phase synchronous generator							
No	Item	Value	Unit	No	Item	Value	Unit
1	Nominal output	9000	kVA	6	Power factor	90/95	
2	Nominal voltage	6300	V	7	Ambient	40	°C
3	Speed	750	rpm	8	Altitude max	1000	m
4	Overspeed	-1388	rpm	9	Voltage range	±10	%
5	Frequency	50	Hz	10	Installed capacity	5.1	MW
Main transformer							
No	Item	Value	Unit	No	Item	Value	Unit
1	Nominal power	9000	kVA	3	Secondary voltage	6.3	kV

2	Primary voltage	110	kV	4	Frequency	50/60 Hz	Hz
Auxiliary transformer							
No	Item	Value	Unit	No	Item	Value	Unit
1	Nominal power	9000	kVA	3	Secondary voltage	400	V
2	Primary voltage	6.3	kV	4	Frequency	50	Hz
Diesel Generator							
No	Item	Value	Unit	No	Item	Value	Unit
1	Component	DG.120		3	Nominal frequency	50	Hz
2	Nominal voltage	3x400	V	4	Nominal power	120 kVA	kVA

1.13 Project Location

The project is under construction in Mok District, Xieng Khouang Province, Lao PDR.

Table 1-1: Project Location

	Latitude	Longitude
Weir 1	19° 0'42.36"	103° 57'55.05"
Weir 2	19° 0'44.09"	103° 58'12.14"
Weir 3(Bath receiver)	19° 0'57.18"	103° 58'27.45"
Power Plants	19° 02'09.4"	103° 59'09.2"

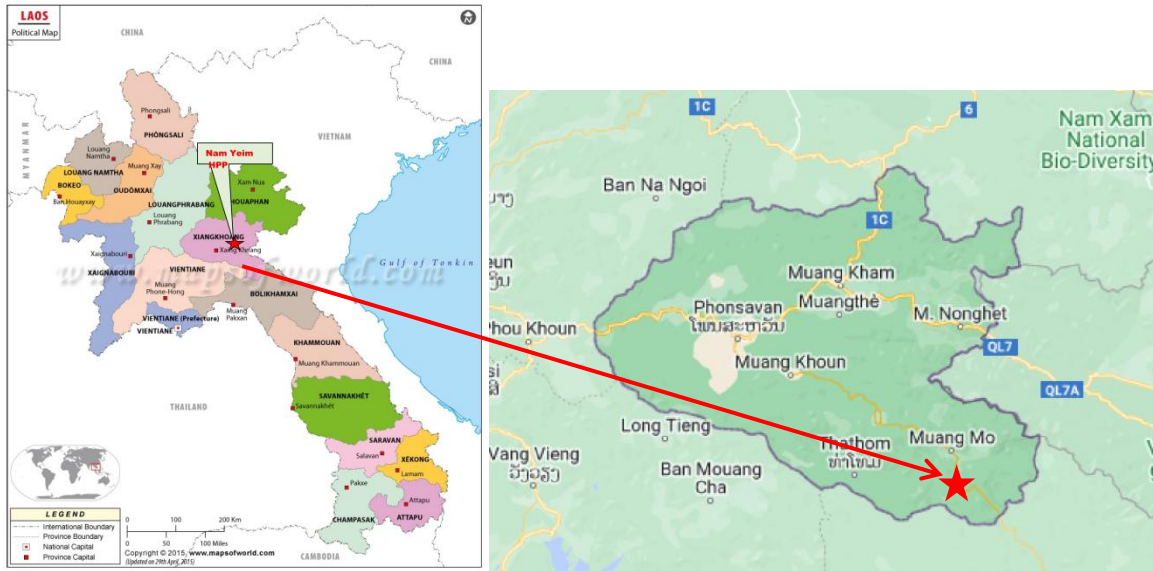


Figure 1-1: Location of Mok District in Lao P

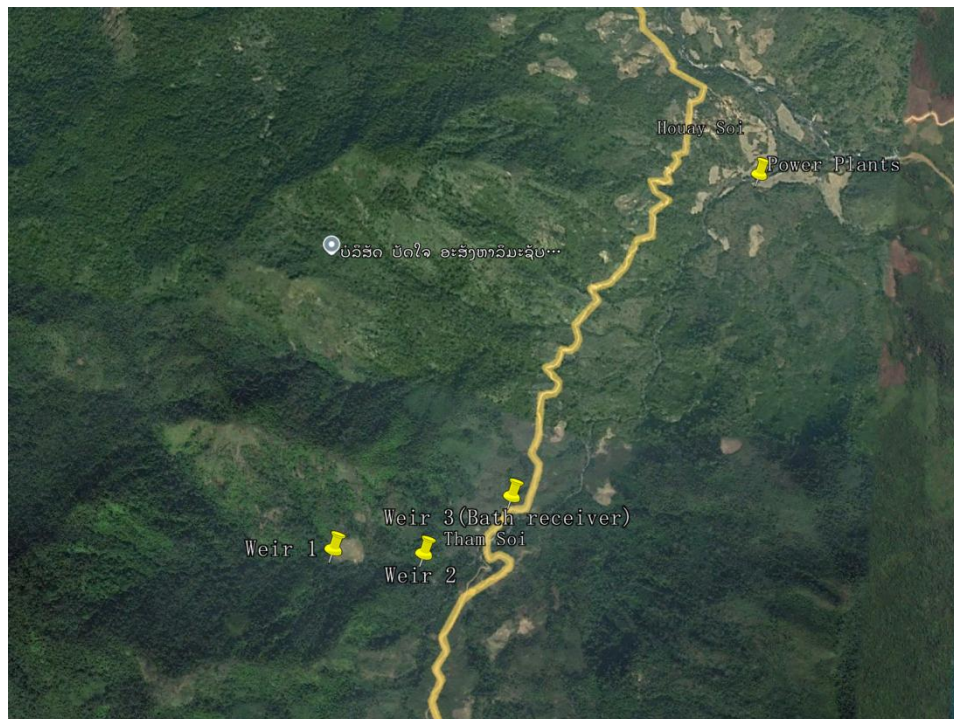


Figure 1-2: Location of the Project site (satellite map)

1.14 Conditions Prior to Project Initiation

The project is a Greenfield hydro power project and does not involve generation of GHG emissions for the purpose of their subsequent reduction, removal or destruction. The project activity generate electricity to national grid of Vietnam and in absence of proposed project the

equivalent amount of electricity would have continued to be generated in grid by fossil fuel plants, which is also the baseline scenario. For more information refer to section 3.4 for the baseline scenario.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project has received all the necessary approvals for the development and commissioning of the proposed 10.2 MW hydro power project from the respective nodal agencies and is in compliance to the local laws and regulations. The list of applicable laws & regulations and the project compliance against it are given below:

Laws & Regulation	Project Compliance
Law on Electricity (amended version) No. 03/NA, dated 20/12/2011 ³ ;	Project received the Approval Certificate for its Feasibility Study Report from the Xieng Khouang Provincial Department of Energy and Mines. This approval confirms the project's compliance with these laws and regulations. The project also signed PPA which is also an approval for electricity connection with the grid.
Decree on the Approval and Promulgation of the Policy on Sustainable Hydropower Development of the Lao PDR No. 02/GoL., dated 12/01/2015 ⁴ ;	
Environmental Protection Law (EPL), No. 29/NA, dated 18/12/2012 ⁵ ;	Project received the Environmental Certificate from the Xieng Khouang Provincial Department of Natural Resources and Environment, based on the submitted Initial Environmental Examination (IEE) Report and the Environmental and Social Management and Monitoring Plan (ESMMP). This certificate confirms the project's compliance with these laws and regulations.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

³<https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf>

⁴policy.asiapacificenergy.org/sites/default/files/Decree%20on%20the%20Approval%20and%20Promulgation%20of%20the%20Policy%20on%20Sustainable%20Hydropower%20Development%20in%20Lao%20PDR.pdf

⁵https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

⁵https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes☒ No

1.18 Sustainable Development Contributions

The project activity implemented by the project owner can contribute to sustainable development as defined by and tracked against the United Nations Sustainable Development Goals (SDGs). The specific analysis is as follows:

SDG 7 : Affordable and Clean Energy

The project contributes SDG Target 7.2.1 “By 2030, increase substantially the share of renewable energy in the global energy mix” by the utilization of hydropower as a renewable energy source. By generating an estimated annual output of 35,000 MWh of renewable energy, this contribution aligns with the overarching goal of fostering a greener and more sustainable future.

Thus, the project will contribute to SDG 7 Affordable and Clean Energy.

SDG 8 : Decent Work and Economic Growth

The project creates direct and indirect employment opportunities during construction and operation phases, so it contributes to SDG Target 8.5.2 “By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities and equal pay for work of equal value”. The project will monitor its contribution to sustainable development by recording the number of permanent jobs created during the operational phase.

Thus, the project will contribute to SDG 8 Decent Work and Economic Growth.

SDG 13 : Climate Action

The project produces clean renewable energy by diminishing CO₂ emissions. Therefore, it contributes SDG Target 13.2.2 “Total greenhouse gas emission per year”. The project activity expected to reduce 22,841 tCO₂e per year.

Thus, the project will contribute to SDG 13 Climate Action.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as no leakage emission is involved in this project activity.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable, as this project activity does not involve any additional information that may have a bearing on the eligibility of the project, the GHG emission reductions, or the quantification of the project's reductions.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

Stakeholders include various individuals, groups, and institutions who have an interest in, could be impacted by the project, possess the ability to significantly influence, or are crucial for achieving the desired project outcome. The list of relevant stakeholders identified through stakeholder mapping includes:

- Representatives of people or groups affected by the proposed project (men, women, minority and marginalized groups).
- Local village head / village chief.
- Local village-level mass organizations
- Local policymakers and representatives of local authorities.

Stakeholders have been invited through various means of invitation like public announcement, invitation letters, etc.

Legal or customary tenure/access rights	Prior to the project implementation, the land within the project boundary was legally classified as state-owned land. While there was no private ownership of the land by local residents, communities in the vicinity historically practiced customary access to specific sections of the project area. These areas were utilized for the collection of wild edibles and natural resources, such as bamboo shoots, mushrooms, herbal medicines, and rattan, which supported local subsistence. The project owner has secured all necessary legal approvals, including the Concession Agreement and Land Use Rights from the relevant government authorities. To address the customary usage, a detailed socio-economic survey was conducted. Acknowledging the economic value of these natural resources to the local households, the project owner has provided fair compensation for the loss of access and potential income, in full compliance with national regulations and the project's resettlement and compensation plan. Following the completion of the compensation process, all customary claims have been effectively settled. The Project Owner currently holds 100% clear legal and customary rights over the project site for the entire crediting period, with no ongoing disputes regarding land tenure or access.
Stakeholder diversity and changes over time	The project is located in Mok District, Xieng Khouang Province. A socio-economic survey was carried out at Mok District, Xieng Khouang Province to determine relevant demographic characteristics of the communities. The project area is characterized by a high degree of cultural homogeneity, predominantly inhabited by the Lao Soung ethnic group. This shared cultural and linguistic background fosters strong social cohesion and reinforces traditional community bonding mechanisms. While a segment of the population consists of civil servants and small business owners, the majority of households rely on an agro-pastoral livelihood system. Within this structure, socio-economic surveys have identified specific households living in poverty. Regarding the workforce, the project employs only a minimal number of foreign workers who serve primarily as technical

	experts. This approach minimizes the risks of social or cultural conflict typically associated with a large influx of external personnel. Conversely, the interaction between the limited number of foreign experts and the broader local workforce facilitates the transfer of technical knowledge to local workers. With the implementation of the project, many farmers are able to diversify their income sources by engaging in labor at the construction sites, while still maintaining their traditional livestock rearing activities. This transition provides a stable cash flow and helps narrow the economic gap between impoverished groups and affluent households, thereby effectively reducing the overall vulnerability of the community. However, to ensure that the project continues to align with the diverse needs and expectations of its stakeholders, the project proponents has implemented a grievance redress mechanism. This mechanism provides a transparent and accessible channel for community members to voice concerns, ensuring that any issues are resolved promptly.
Expected changes in well-being	The implementation of the project is expected to have a positive impact on the well-being and other characteristics of stakeholders. Relative to the baseline scenario, the project will bring about the following changes: Climate change Impact By replacing electricity generated from fossil fuels on the Vietnamese grid with renewable power, the project reduces GHG emissions. Social welfare During the construction and operation of the project, it would bring a large number of job opportunities to the local people, especially those from poor families. Then it will help them earn income through working in the project, thus contributing to the economic stability of the local community. Collectively, these changes will have a positive impact on residents' livelihoods and living environment, while enhance the sustainability of local communities.
Location of	The location of the stakeholders is Mok District, Xieng Khouang

2.1.2 Stakeholder Consultation and Ongoing Communication

19

	1.	Gender	Male	26
			Female	24
	2.	Age	≤25	8
			26~54	32
			≥55	10
	3.	Educational background	≤Junior High School	28
			High school and junior college	12
			Bachelor	9
			Master	1
	4.	Identity	Resident of the project area	30
			Local community representative / Village leader	4
			Staff of local government department	5
			Employee of the project investor/developer	10
			Other (Please specify)	1
	Consultation outcome			
	In addition, FPIC (Free, prior, and informed consent) was discussed to ensure the legitimacy and transparency of the project. The project also discusses the VCS validation process to ensure the sustainability and environmental benefits of the project. Based on the results of the discussion at the meeting and the 50 questionnaires received for the survey, opinions are summarized as follows:			
	No.	Question	Answer	Amount
1	Prior to this survey, how familiar were you with this	very well known	20	

		hydro power project?	relatively understand	28
			Don't know	2
	2	Do you think that the construction of the project will be beneficial to the local environment?	Yes	42
			No	0
			Don't know	8
	3	How do you think the construction of this project will affect the region?	Improve local infrastructure	22
			Provide employment opportunities	28
			Destroy the surrounding environment	0
	4	Do you think the construction of this project will be beneficial to the economic development of the region?	Yes	44
			No	0
			Don't know	6
	5	Do you think that the construction of this project will have any adverse effects on your life, studies or work?	Yes	0
			No	44
			Don't know	6
	6	Do you support the construction of this project?	Yes	50
			No	0
			Not care	0
Ongoing communication		The project proponent has a grievance mechanism. Stakeholders can anytime provide their inputs or concerns over the project		

	through any of the following means: • Grievance register: Stakeholders can provide the grievance in writing in the grievance register maintained at the site. • Telephone access: Stakeholders can provide their grievances through calling the designated site telephone number. • Email access: Stakeholders can provide their grievances through email using the official project email address. The site manager is responsible for addressing the grievances received from stakeholders. The same is informed during the stakeholder consultation meeting. This mechanism will ensure to continue to engage with local stakeholder in an ongoing proactive manner.
Stakeholder input	During the consultation meeting, the village head raised specific requests regarding community infrastructure, specifically asking for the project developer's assistance in constructing a new village administration office and upgrading local access roads. Due account was taken of this input. While these requests do not necessitate changes to the technical engineering design of the hydro power plant, they have been integrated into the project's community development framework. The project owner has formally acknowledged these needs and committed to providing support commensurate with its actual capabilities and resource availability during the construction and operation phases.

2.1.3 Free Prior and Informed Consent

Obtaining consent	The project activity has been implemented with all necessary approvals from the Department of Mines and Energy and the Department of Natural Resources and Environment of Xiangkhouang Province. Project proponent has conducted LSC during which all the necessary information regarding the project activity was explained to the stakeholders. The project proponent fostered open dialogues, providing stakeholders ample opportunities to express concerns, ask questions, and share input. All raised concerns and inputs have been documented, and a grievance resolution mechanism is in place to address any ongoing conflicts, demonstrating the project's commitment to prioritizing resolution and community harmony.
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	Regular stakeholder engagement will persist throughout the project lifecycle to address evolving concerns and ensure ongoing support. Through this comprehensive process, the project not only obtained consent but also built a foundation of trust and cooperation with stakeholders, resulting in no ongoing or unresolved conflicts.
Outcome of FPIC	The result of the FPIC and stakeholder consultation is positive, with stakeholders expressing their support for the project. Project proponent has ensured that the project did not encroach on land, relocate people without consent, or cause forced physical or economic Displacement.

2.1.4 Grievance Redress Procedure

Development process	The grievance procedure was discussed during local stakeholder meeting. As, discussed the project owner shall maintain a grievance register at site, wherein any stakeholder can provide feedback/inputs if any concerns. The project owner has provided email IDs of concerned persons at site and for escalation if needed. Any concern raised will be initially taken up by team at site, however if no satisfactory solutions provided the stakeholder can escalate the same to General Manager, the concerns raised will be taken and resolved within agreed time frame as reasonable.
Grievance redress procedure	The project proponent has established a record keeping system and if any feedback/complain received shall be recorded and remedy to same will be done with information to concern person/entity. If any concern/feedback received will be discussed and if require stakeholder will be invited again to discuss and take necessary action as may be needed.

2.1.5 Public Comments

Comments received	Actions taken
To be added	To be added

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The project has established a professional management team equipped with the experience and expertise required to effectively implement similar project activities. Team members possess years of working experience in the hydropower and environmental protection sectors, along with specific expertise in project planning, implementation, and supervision, enabling them to effectively manage and advance the project. Furthermore, the project owner has extensive experience in operating hydropower projects, ensuring that the project is managed in accordance with high industry standards.

In addition, a recruitment strategy has been developed to fill potential gaps in skills and experience within the team. Professionals with relevant experience and expertise will be recruited to ensure that the project is effectively supported and guided.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	The implementation of the project and its impact on the natural environment comply with relevant laws and regulations and are expected to have a positive impact on the wellbeing of stakeholders. Therefore, the project has no natural and human induced risks to stakeholders' wellbeing.
Risks to stakeholder participation	No risk identified	The stakeholders were provided with brief information related to project and agenda before stakeholder consultation process and there were no unresolved concern/conflict with any relevant stakeholders to the project. Hence, no potential risk identified which may impact stakeholder participation to the project.
Working conditions	No risk identified	Occupational Health and Safety will be prioritized at every stage of the project. The project implements "Zero Accident" program, daily Toolbox Meetings, provide mandatory PPE (earplugs, etc.), and conduct annual health checks. The project is committed to complying with national Occupational Health and Safety Standards ⁶ , national labour standards ⁷ , IFC

⁶ https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=73666

		PS2 ⁸ and International Labour Organization Labor Standards, following all guidelines and standards.
Safety of women and girls	No risk identified	The project promotes safe and secure environments for all stakeholders including women and girls. During the implementation of the project, all personnel involved in the project are regularly propagate in gender sensitivity, including awareness of gender discrimination and gender-based violence, and how to prevent and respond to these issues. The project has established effective monitoring and reporting mechanisms so that women can report any safety issues or acts of gender discrimination, ensuring that their voices are adequately addressed and protected.
Safety of minority and marginalized groups, including children	No risk identified	The project strictly complies with national legislation that explicitly forbids both forced labour and child labour. The project provides equal job opportunities, does not discriminate against marginalized groups, and tries to provide some social support and assistance for local marginalized groups. Therefore, the project does not impact minority and marginalized groups including children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project is a hydro power project, hence the project will not result in release of pollutants to the environment during the operation phase. The project has a temporary impact on the environment only during the construction phase due to soil disruption, particulate dust generation and noise associated with vehicles, respective measures have been taken to minimize the impact.

2.3 Respect for Human Rights and Equity

⁷ https://natlex.ilo.org/dyn/natlex2/r/natlex/fe/details?p3_isn=96369

⁸ <https://www.ifc.org/content/dam/ifc/doc/2010/2012-ifc-performance-standard-2-en.pdf>

2.3.1 Labor and Work

	Risks identified ⁹	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	The project does not involve or engage in any form of discrimination based on gender, race, religion, sexual orientation, or any other grounds. This commitment underscores the project's dedication to promoting equality and inclusivity by refraining from practices that may unfairly disadvantage any individual or group.
Sexual harassment	No risk identified	The project emphasizes its commitment to creating a safe and respectful work environment free from sexual harassment. Since its inception, there have been no complaints of sexual harassment. Moreover, the project has proactively established a monitoring mechanism specifically designed to address complaints about sexual harassment. This mechanism ensures that any instances of sexual harassment are promptly investigated and addressed, thereby preventing their occurrence in the future and reinforcing the project's dedication to maintaining a harassment-free workplace.
Equal pay for equal work	No risk identified	The project implements equal pay for equal work to ensure that workers' remuneration is commensurate with the amount of work done and their performance. Each position is regularly evaluated to determine the nature, difficulty, and volume of work, as well as the skills required. A corresponding remuneration system is formulated based on the results of these job evaluations to ensure that laborers engaged in the same or similar work receive the same labor compensation.
Gender equity in labor and work	No risk identified	The project has male and female employees of different genders. There is no significant

⁹ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		gender difference between female employees and male employees in terms of salary, promotion opportunities and welfare benefits, and employees of different genders have equal opportunities to obtain training and development resources. These policies ensure that all employees are treated equally in terms of gender and pay.
Forced labor	No risk identified	Project proponent adheres to international labor standards, particularly those set by the International Labor Organization (ILO), to ensure that its recruitment and employment policies do not involve any form of forced labor. Project proponent conducts comprehensive background checks on all employees and complies with local labor laws and regulations to prevent any exploitation or abuse. Additionally, the project has implemented a supervision mechanism, including regular on-site inspections and audits, to identify and immediately address any potential violations of labor rights, including forced labor.
Child labor	No risk identified	In this project, child labor is unacceptable and strictly prohibited. The project adheres to the relevant standards of the International Labour Organization as well as the local laws and regulations that prohibit the employment of minors. At the same time, the project has established a supervision mechanism and conducts regular on-site inspections and audits to ensure that there is no child labor in our operations. Project proponent also provides training for employees and partners to enhance their awareness of the harm caused by child labor and encourages them to report any suspicious situations.
Human trafficking	No risk identified	The project firmly opposes human trafficking. The project has a compliant recruitment and employment policy that complies with the standards of the International Labour Organization and does not involve any form of

		human trafficking. At the same time, it conducts strict background checks on all employees and partners to identify and prevent any possible acts related to human trafficking. The project also implements supervision mechanisms, including regular on-site inspections and audits, to ensure that there is no human trafficking in the business and that the rights of all individuals are respected. Project proponent also provides training for employees and partners to enhance their understanding of human trafficking and educate them on how to avoid and report possible violations.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	The project adheres to human rights principles, ensuring the protection of all stakeholders. The project promotes gender equality, prevents discrimination, and prohibits child and forced labor. It follows national labor laws and international standards, including the Universal Declaration of Human Rights and ILO conventions. Equal pay for equal work is enforced, and all workers are provided with safe and respectful working conditions. The project is committed to fostering an inclusive environment for women, marginalized groups, and vulnerable populations, ensuring their rights are protected throughout the project lifecycle.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified.	This project utilizes hydro power and creates

employment opportunities for skilled and unskilled workers. The project owner has secured legal land use rights; to address historical customary usage for natural resource collection, fair compensation was provided to ensure no adverse impact on stakeholders. The proponent remains committed to upholding indigenous rights through transparency, ensuring positive impacts on the ecosystem and local livelihoods.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	This project utilizes hydro power. During the implementation and operation phases, the project activities created employment opportunities for skilled and unskilled workers. The project owner has formally obtained the land use rights for the project area. Consequently, no property or customary rights of any stakeholders are infringed upon. The project proponent is committed to upholding the rights of indigenous peoples by ensuring transparency in project planning and execution, ultimately delivering positive impacts on nearby residents through an improved ecosystem and employment opportunities.

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	Not applicable as the project owner has formally obtained the land use rights for the project area. There, is no explicit benefit sharing arrangement with the local stakeholders of any kind neither is any such arrangement required as per local laws.
Summary of the benefit sharing plan	Not applicable.

Approval and dissemination of benefit sharing plan	Not applicable.
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2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	This project is of the run-of-the-river type and the construction process does not involve deep well drilling. Hence, groundwater is not expected to be impacted during construction. During the operation phase, the water flow adheres to the natural course of the river downstream, ensuring that water utilization does not impact groundwater resources. The monitoring plan stipulates that groundwater levels at the project site be measured regularly during the operation phase. Furthermore, desilting of the reservoir area will be conducted every 3-5 years to ensure optimal water storage capacity.
Soil degradation and soil erosion	No risk identified	There is no negative impact due to project instances on soil. Moreover, the project plans plantation activity near project sites to prevent any potential soil erosion.
Water consumption and stress	No risk identified	There are no competing uses for water resources in the Project locations. As the project is a run-of-the-river project, it will divert water stream for short distance and onwards releases the same to main river stream; this has no impact on water availability and does not create water stress. Furthermore, in line with the conservative practices of design, minimum discharge is maintained in river stream, which shall meet any other water requirements in the region for other social needs.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes ☐ No

2.4.2 Introduction of Species

The project activity is not a AFOLU project,hence not applicable.

2.4.3 Ecosystem Conversion

The project activity is not a AFOLU project,hence not applicable.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	AMS-I.D.	Grid connected renewable electricity generation	18.0
Tool	Tool 7	Tool to calculate the emission factor for an electricity system	7.0
Tool	Tool 21	Demonstration of additionality of small-scale project activities	13.1
Tool	Tool 27	Investment Analysis	14.0

3.2 Applicability of Methodology

The project activity involves electricity generation using renewable hydro energy with a total installed capacity of 10.2 MW. As the capacity is below the 15 MW threshold, the project qualifies as a Type I small-scale project activity. The selected methodology and tools, and their applicability to the project, are discussed below.

Methodology ID	Applicability condition	Justification of compliance
AMS-I.D.	This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind,	The project activity is hydro power generation and supplies electricity to the

	geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	Vietnam national grid, falling under condition (a). Hence, this criterion is met.
AMS-I.D.	Illustration of respective situations under which each of the methodology (i.e. "AMS-I.D.: Grid connected renewable electricity generation " , " AMS-I.F.: Renewable electricity generation for captive use and mini-grid " and " AMS-I.A.: Electricity generation by the user) applies is included in the appendix.	The project supplies electricity to the Vietnam national grid, satisfying the first point in the appendix table. Hence, this criterion is met.
AMS-I.D.	This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	The project is a newly built grid-connected hydro power plant, falling under condition(a). Hence, this criterion is met.
AMS-I.D.	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of	This project results in new reservoirs. As calculated in Section 4.2, the power density of the project activity is 3319.77W/m², which is significantly greater than 4W/m². Hence, this criterion is met.

	reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m²; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m².	
AMS-I.D.	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	The proposed project is a 10.2 MW hydro power project equipped with one diesel generator as an emergency backup power source. The project meets the eligibility limit of 15 MW for the renewable component. Hence, this criterion is met.
AMS-I.D.	Combined heat and power (co-generation) systems are not eligible under this category.	This is not applicable to the project activity as the project involves only hydroelectric generating units and does not involve heat generation units.
AMS-I.D.	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct¹⁰ from the existing units.	This is not applicable to the project activity as the project is a greenfield project and does not involve the capacity addition of renewable energy generation units at an existing renewable power generation facility.

¹⁰ Physically distinct units are those that are capable of generating electricity without the operation of existing units, and that do not directly affect the mechanical, thermal, or electrical characteristics of the existing facility. For example, the addition of a steam turbine to an existing combustion turbine to create a combined cycle unit would not be considered "physically distinct".

AMS-I.D.	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	This is not applicable to the project activity as the project is a greenfield project and does not involve the retrofit, rehabilitation or replacement of existing renewable energy power generation facilities.
AMS-I.D.	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as " AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	This is not applicable to the project activity as the project is a hydro power project, not a landfill gas, waste gas, wastewater treatment, or agro-industries project.
AMS-I.D.	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	This is not applicable to the project activity as the project is a hydro power project and does not involve biomass.
Tool 7	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project applies the OM, BM, and CM values published in the "Viet Nam national grid emission factor report 2023" (attached with the Official Letter No. 1726/BDKH-PTCBT published on 12 March 2024). All three values in this report were calculated in accordance with Tool7.

		Please refer to Section 4.1 for the detailed calculation process. Hence, the tool is applicable.
Tool 7	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in " Appendix 1: Procedures related to off-grid power generation " should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Since the project supplies power to the national power grid of Vietnam, tool7 has been used to calculate the emission factors of the power grid. Hence, the tool is applicable.
Tool 7	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Lao PDR and generated electricity supplied to Vietnam grid, both are non-Annex I countries. Hence, this tool is applicable.
Tool 7	Under this tool, the value applied to the CO₂ emission factor of biofuels is zero.	This is not applicable to the project activity as the project does not involve biofuels usage.

Tool 21	The use of the methodological tool "Demonstration of additionality of small-scale project activities" is not mandatory for project participants when proposing new methodologies. Project participants and coordinating/managing entities may propose alternative methods to demonstrate additionality for consideration by the Executive Board.	The project proponent is not proposing any new methodology and has applied this tool for the demonstration additionality in compliance with the tool. For details, please refer to Section 3.5 . Hence, this tool is applicable.
Tool 21	Project participants and coordinating/managing entities may also apply " TOOL19: Demonstration of additionality of microscale project activities" as applicable.	This is not applicable to the project activity as the project is not a microscale project.
Tool 27	This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	The investment analysis has been conducted in accordance with Tool 27. For details, please refer to Section 3.5.2. Hence, the tool is applicable.
Tool 27	In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The investment analysis method selected by the project is consistent with the requirements of Tool 27. Please refer to Section 3.5.2 for the detailed analysis. Hence, the tool is applicable.

3.3 Project Boundary

This section will be completed in the full Project Description(PD) submitted for the ' under validation' status, as this version of the PD is for pipeline listing purposes only.

Source		Gas	Included?	Justification/Explanation
Baseline	Source1	CO ₂		
		CH ₄		

Source	Gas	Included?	Justification/Explanation
	N ₂ O		
Project	Source2		
	CO ₂		
	CH ₄		
	N ₂ O		

3.4 Baseline Scenario

This section will be completed in the full Project Description(PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

The project activity is not enforced by any law, statute, or regulatory framework in Lao PDR.

The project passes the regulatory surplus test as there are no enforced laws, regulations, court orders, or environmental mitigation agreements that mandate the implementation of this specific hydro power project. Any concession agreements or power purchase agreements signed by the project owner are voluntary commercial commitments and do not constitute a legal mandate for the purpose of additionality.

While mandatory laws and regulations exist in the sector, they regulate how the project is constructed and operated, not whether it must be implemented. These include:

Law on Electricity (2012)¹¹

¹¹ <https://policy.asiapacificenergy.org/sites/default/files/Electricity%20Law.pdf>

Environmental Protection Law (2012)¹²

None of the above-referenced regulations mandate the implementation of hydro power project.

3.5.2 Additionality Methods

This section will be completed in the full Project Description(PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.

3.6 Methodology Deviations

There is no methodology deviation.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

According to methodology AMS-I.D (Version 18.0), baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,y}$$

Where:

BE_y	=	Baseline emissions in year y (t CO ₂)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)
$EF_{grid,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO ₂ /MWh)

Calculation of $EG_{PJ,y}$

¹²

https://www.ajcsd.org/chrip_search/dt/pdf/Useful_materials/Lao_PDR/Environmental_Protection_Law_Revised_Version.pdf

Since the project activity consists of the installation of new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, the project activity is the installation of a greenfield renewable energy power plant, so that:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant to the grid in year y (MWh)

Calculation of $EF_{grid,y}$

The emission factor should be calculated in a transparent and conservative manner according to the procedures prescribed in the Tool7 “Tool to calculate the emission factor for an electricity system” (Version 07.0).

The data used for calculation are from an official source (where available) and publicly available. The calculation processes are as follows:

Step1: Identify the relevant electricity system;

Step2: Choose whether to include off-grid power plants in the project electricity system (optional);

Step3: Select a method to determine the operating margin (OM);

Step4: Calculate the operating margin emission factor according to the selected method;

Step5: Calculate the build margin (BM) emission factor;

Step6: Calculate the combined margin (CM) emissions factor.

Step1: Identify the relevant electricity system

This project will be connected to the national electricity grid of Viet Nam, which is operated and monopolized by the EVN. This national electricity grid is the unique transmission and distribution line, to which all power plants in Vietnam are physically connected. Hence, the national electricity grid is the project electricity system.

There are electricity imports to the Viet Nam national electricity grid from China and Laos, thus the China and Laos Power Grid is the connected electricity systems. As per Tool 07, for the purpose of determining the operating margin emission factor, the CO₂ emission factor for net electricity imports from connected electricity systems is 0 tCO₂/MWh by default.

Step2: Choose whether to include off-grid power plants in the project electricity system (optional)

There are two options to calculate the operating margin and build margin emission factor.

Option1: Only grid power plants are included in the calculation;

Option 2: Both grid power plants and off-grid power plants are included in the calculation

Because only the data of grid connected power plants is available, so Option I will be chosen for calculating the grid emission factor.

Step3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step4:

(a) Simple OM; or

(b) Simple adjusted OM; or

(c) Dispatch data analysis OM; or

(d) Average OM.

The method (a) “Simple OM” was chosen for Viet Nam because the low cost/must-run resources constitute less than 50% of the total grid generation in the most recent five years (detail see in the table below):

Table 3-1: Rate of low cost/must-run resources based on generation.

Year	2019	2020	2021	2022	2023	total
Hydropower	54,411,106	59,387,446	69,606,845	78,225,090	70,469,947	333,702,417
Bagasse power	280,996	331,319	347,560	246,621	868,000	2,074,496
Wind power	721,189	946,157	3,243,227	8,902,337	12,481,822	26,414,247
Solar power	4,833,674	9,684,525	15,141,520	14,381,800	16,083,045	60,124,564
Import	3,316,000	3,067,000	1,401,463	1,357,845	4,220,000	15,275,308
Total grid generation	207,214,694	207,692,796	208,561,267	220,446,789	236,285,221	/
Low-cost/Must-run ratio						40.51%

The Ex-ante option was chosen, thus, no monitoring and recalculation of the emissions factor during the crediting period is required. The 3-year generation-weighted average (2021, 2022 and 2023) that is the most recent data available was used to calculate the OM emission factor.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit;
Or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

The Option B was chosen by DNA of Viet Nam, “Viet Nam national grid emission factor report 2023” (attached with the Official Letter No. 1726/BDKH-PTCBT” published on 3/12/2024, in line with Tool 07 because the necessary data for Option B is available. Under Option B, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low cost/must run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \sum (FC_{i,y} \times NCV_{i,y} \times EF_{CO_2,i,y}) / EG_y$$

Where,

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO_2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ /GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

On the basis of the data available, the three-year average operating margin emission factor is calculated by the DNA as a full-generation-weighted average of the emission factors:

$$EF_{grid,OM,y} = 0.9201 \text{ tCO}_2/\text{MWh}$$

Step 5: Calculate the build margin emission factor ($EF_{BM,y}$)

In terms of vintage of data, project participants can choose between one of the following two options:

Option1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

Option2: For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The Option 1 was chosen by DNA of Viet Nam, “Viet Nam national grid emission factor report 2023” (attached with the Official Letter No. 1726/BDKH-PTCBT” published on 3/12/2024. The BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5-units}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);

The set of five most recently built power units ($SET_{5-units}$) is indicated in Appendix X. These five power units generated 220,141.6 MWh of electricity ($AGE_{SET-5-units}$) in 2023.

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET_{\geq 20\%}}$, in MWh);

The total output of Viet Nam electricity grid (AEG_{total}) in 2023 is 47,402,236.48 MWh. The most recent-built power plants ($SET_{\geq 20}$ per cent) selected by DNA of Viet Nam accounted for 20.06% of the system electricity generation (AEG_{total}) and generated 236,285,220.98 MWh of electricity ($AEG_{SET_{\geq 20\%}}$) in 2023.

(c) From $SET_{5\text{-units}}$ and $SET_{\geq 20 \text{ percent}}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample}).

$SET_{\geq 20 \text{ percent}}$ (236,285,220.98 MWh) is larger than $SET_{5\text{-units}}$ (220,141.6 MWh); and none of the power units in $SET_{\geq 20 \text{ percent}}$ started to supply electricity to the grid more than 10 years ago, then $SET_{\geq 20 \text{ percent}}$ is chosen as SET_{sample} and use SET_{sample} to calculate the build margin.

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available. It is calculated as follows:

$$EF_{grid, BM, y} = \frac{\sum_m EG_{m, y} \times EF_{EL, m, y}}{\sum_m EG_{m, y}}$$

Where,

$EF_{grid, BM, y}$	=	Build margin CO_2 emission factor in year y ($t CO_2/MWh$)
$EG_{m, y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL, m, y}$	=	CO_2 emission factor of power unit m in year y (tCO_2/MWh)
m	=	Power units included in the build margin
y	=	Most recent historical year for which power generation data is available

$$EF_{grid, BM, y} = 0.3983 \text{ tCO}_2/\text{MWh}$$

Step 6: Calculate the combined margin emissions factor

According to Tool 07, the calculation of the combined margin (CM) emission factor ($EF_{grid, CM, y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM

The weighted average CM method (option a) is the preferred option and used to calculate the CM emission factor of the national grid.

The combined margin emissions factor is calculated as follows:

$$EF_{grid, CM, y} = EF_{grid, OM, y} \times \omega_{OM} + EF_{grid, BM, y} \times \omega_{BM}$$

Where,

$EF_{grid, BM, y}$	=	Build margin CO_2 emission factor in year y ($t CO_2/MWh$)
$EF_{grid, OM, y}$	=	Operating margin CO_2 emission factor in year y ($t CO_2/MWh$)

ω_{OM} = Weighting of operating margin emissions factor (per cent)

ω_{BM} = Weighting of build margin emissions factor (per cent)

Except for wind and solar power generation project activities, the operating margin and build margin for all other projects are 50% and 50%. As the project activity is hydro power project the above weight is applied to calculated combined margin emission factor.

Therefore, the baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor ($EF_{OM,y}$) and the Build Margin emission factor ($EF_{BM,y}$) as below:

$$EF_{grid,CM,y} = 0.9201 \times 0.5 + 0.3983 \times 0.5 = 0.6592 \text{ tCO}_2/\text{MWh}$$

Hence the baseline emission is calculated as below:

$$BE_y = 34,650 \times 0.6592 = 22,841 \text{ tCO}_2$$

4.2 Project Emissions

The project activity is a hydro power project activity and as per the para 39 of the applied methodology the Emissions from water reservoirs of hydro power plants shall be determined considered following the procedure described in the most recent version of "ACM0002: Grid connected electricity generation from renewable sources". The project emissions are to be calculated as follows:

$$PE_y = PE_{FF,y} + PE_{HP,y} + PE_{PSP,y}$$

Where,

PE_y = Project emissions in year y (t CO₂e/yr)

$PE_{FF,y}$ = Project emissions from fossil fuel consumption in year y (t CO₂/yr)

$PE_{HP,y}$ = Project emissions from water reservoirs of hydro power plants and pumped storage plants in year y (t CO₂e/yr)

$PE_{PSP,y}$ = Project emissions from utilizing electricity from the grid for pumping operation of PSP in excess to the production of the renewable power plant operating in coordination with the PSP (t CO₂e/yr)

Emissions from fossil fuel combustion($PE_{FF,y}$)

As per the para 39 of the applied methodology, all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.

Emissions from water reservoirs of hydro power plants and pumped storage projects ($PE_{HP,y}$)

The power density (PD) of the project activity is calculated as follows:

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where,

PD	=	Power density of the project activity (W/m ²)
Cap _{PJ}	=	Installed capacity of the hydro power plant or PSP after the implementation of the project activity (W)
Cap _{BL}	=	Installed capacity of the hydro power plant or PSP before the implementation of the project activity (W). For new hydro power plants or PSP, this value is zero.
A _{PJ}	=	Area of the single or multiple reservoirs measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m ²)
A _{BL}	=	Area of the single or multiple reservoirs measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m ²). For new reservoirs, this value is zero.

$$PD = (10,200,000-0)/(3072.5-0) = 3319.77\text{W/m}^2$$

As per ACM0002 (Version 22.0), If the power density of the project activity is greater than 10 W/m², PE_{HP,y} = 0.

Emissions from utilizing grid electricity by pumped hydro plants (PE_{PSP,y})

Since the project does not consume electricity from the power grid, PE_{PSP,y} = 0.

In conclusion, the project emission of the project activity is 0.

4.3 Leakage Emissions

As per applied approved methodology no leakage emission relevant to project activity. LE_y = 0.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

The ex-ante emission reductions (ER_y) for the project activity are calculated as follows:

$$ER_y = BE_y - PE_y - LE_y$$

Where,

ER _y	=	Emission reductions in year y (t CO ₂)
BE _y	=	Baseline Emissions in year y (t CO ₂)
PE _y	=	Project emissions in year y (t CO ₂)
LE _y	=	Leakage emissions in year y (t CO ₂)

Emissions reductions are calculated as:

$$ER_y = 22,841 \text{ tCO}_2\text{e/yr} - 0 \text{ tCO}_2\text{e/yr} - 0 \text{ tCO}_2\text{e/yr} = 22,841 \text{ tCO}_2\text{e/year}$$

The emission reductions can be obtained as below:

Table 5-1 The estimation of ER_y

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
01-Mar-2026 to 31-Dec-2026	19,149	0	0	19,149	0	19,149
01-Jan-2027 to 31-Dec-2027	22,841	0	0	22,841	0	22,841
01-Jan-2028 to 31-Dec-2028	22,841	0	0	22,841	0	22,841
01-Jan-2029 to 31-Dec-2029	22,841	0	0	22,841	0	22,841
01-Jan-2030 to 31-Dec-2030	22,841	0	0	22,841	0	22,841
01-Jan-2031 to 31-Dec-2031	22,841	0	0	22,841	0	22,841
01-Jan-2032 to 31-Dec-2032	22,841	0	0	22,841	0	22,841
01-Jan-2033 to 31-Dec-2033	22,841	0	0	22,841	0	22,841
01-Jan-2034 to 31-Dec-2034	22,841	0	0	22,841	0	22,841
01-Jan-	22,841	0	0	22,841	0	22,841

2035 to 31-Dec- 2035						
01-Jan- 2035 to 14-Jan- 2036	3,692	0	0	3,692	0	3,692
Total	228,410	0	0	228,410	0	228,410

5 MONITORING

5.1 Data and Parameters Available at Validation

This section will be completed in the full Project Description(PD) submitted for the ‘under validation’ status, as this version of the PD is for pipeline listing purposes only.

Data / Parameter	
Data unit	
Description	
Source of data	
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	
Comments	

5.2 Data and Parameters Monitored

This section will be completed in the full Project Description(PD) submitted for the ‘under validation’ status, as this version of the PD is for pipeline listing purposes only.

Data / Parameter	
Data unit	

Description	
Source of data	
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	
Value applied	
Monitoring equipment	
QA/QC procedures to be applied	
Purpose of data	
Calculation method	
Comments	

5.3 Monitoring Plan

This section will be completed in the full Project Description(PD) submitted for the 'under validation' status, as this version of the PD is for pipeline listing purposes only.